**Taibah University**
**College of Computer Science and Engineering**
**Department of Cybersecurity**

# CS372 Database System

## Project

## "Online Retail Application Database System"

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# 1. Introduction

## 1.1 Problem description

Online retail application refers to buying and selling products and services through electronic platforms over the internet.

Although the significant growth in e-commerce in recent years, it still face several challenges due to the massive amount of users and data it processes such as product details, customer information, and sensitive financial data.

Accurately managing and storing these information can be complicated, potentially resulting in problems like payment errors, create an incorrect invoice, challenges in tracking orders, and a decline in data integrity and consistency.

Furthermore, system performance might suffer under heavy loads, leading to a slow response times that adversely affect the user experience.

## 1.2 proposed solution

The proposed solution is a database model that covers all stages of the process, including customer registration, bank account management, invoicing, and other steps.
This database model aims to provide a comprehensive picture of all components in an organized and relational manner, ensuring streamlined purchasing workflows and accurate tracking between entities, such as customers, orders, items, bank accounts, invoices, and suppliers, while clarifying the relationships between them.

## 2. Data requirement and conceptual Model Design

### 2.1 Entities and attributes

**Entities**: A rectangular shape containing the entity name

**Attributes**: An oval shape connected to the entity, with the primary key identified by a line under the name.

| Entities | Description | Attributes | key attributes |
|---|---|---|---|
| **Customer** | Represents the account information used by the customer to place orders and contains unique login credentials | Customer ID<br>UserID<br>Password | Customer ID |
| **BankAccount** | This entity specifies the bank account details associated with the customer | BankName<br>AccountNumber | Account Number |
| **Order** | Represents orders created by customers that contain multiple items | Order ID<br>Order Date<br>Quantity | Order ID |
| **Supplier** | This entity identifies the suppliers who supply the products | SupplierID<br>SupplierName | SupplierID |
| **Bill** | Represents the final invoice generated after the order is placed | Bill ID<br>Total Price | Bill ID |
| **Items** | Represents the goods or products available for purchase | ItemID<br>ItemName<br>Price | ItemID |

### 2.2 Relationships

**Relationships**: A specific shape connecting entities.

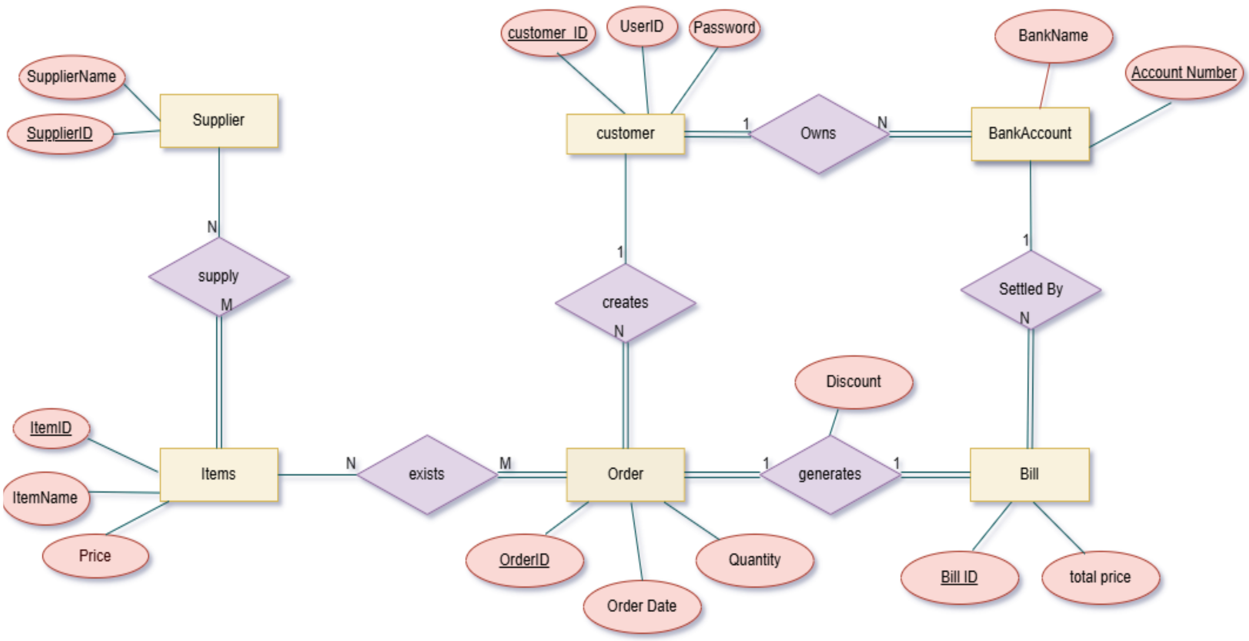
**Cardinality**: Determined by the numbers placed next to each entity on the line connecting the relationship.

**Total Participation**: Represented by a double line between the entity and the relationship.

**Partial Participation**: Represented by a single line.

| Relationships | Between | Cardinality | Total and Partial Participation |
|---|---|---|---|
| **Owns** | Customer & BankAccount | **1:N** Means that a single customer can have more than one bank account. | **BankAccount**: total<br>**Customer**: total<br>Every customer must own a bank account, and every bank account must be owned by a customer. |
| **Creates** | Customer & Order | **1:N** Means that a single customer can create more than one order. | **Customer** :partial<br>Means that not every customer must have created an order.<br>**Order**: total<br>Means that each order must be created by at least one customer. |
| **exists** | Order & Items | **N:M** Means that a set of items can exist within a set of orders. | **Order**:total<br>Means that each order must contain at least one item.<br>**Items**:partial<br>Means that an item can exist without being part of any order. |
| **Generates (Discount)** | Order & Bill | **1:1** Means that one order generates only one invoice. | **Order:total**<br>Means that each order must generate at least one bill.<br>**Bill:total**<br>Means that each bill must be generated by at least one order. |
| **Settled By** | Bill & BankAccount | N:1<br>Means that more than one invoice can be settled from a single bank account. | **Bill**:total<br>Means that each invoice must be settled by at least one account.<br>**BankAccount**:partial<br>Means that there may be a bank account that has not been used to settle any invoice. |
| **Supply** | Supplier & Items | **N:M**<br>Means that a supplier can supply more than one item, and an item can come from more than one supplier. | **Supplier**:partial<br>Means that not every supplier must supply items.<br>**Items**: total<br>Means that each item must be supplied by at least one supplier. |

## 2.3 EER Diagram




# 3. Relational Model

## 3.1 Relational Schema

**Customer**

| | | | |
|---|---|---|---|
| Customer | Customer_ID | UserID | Password |

**Bank_Account**

| | | | |
|---|---|---|---|
| BankAccount | AccountNumber | BankName | Customer_ID |

**OrderWithBill**

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Order | OrderID | OrderDate | Quantity | Customer_ID | Bill | BillID | TotalPrice | AccountNumber | Discount |

**Item**

| | | | |
|---|---|---|---|
| Items | ItemID | ItemName | Price |

**Supplier**

| | | |
|---|---|---|
| Supplier | SupplierID | SupplierName |

**Exists**

| | |
|---|---|
| OrderID | ItemID |

**Supply**

| | |
|---|---|
| SupplierID | ItemID |

## 3.2 Details of mapping EER into a Relational Schema

<table>
<tr><th>Steps</th><th>Description</th></tr>
<tr><td>Step1: Mapping regular entity types</td><td>Each regular entity was mapped into a separate relational table.<br><br>The following tables are created, along with their corresponding primary keys: Customer (CustomerID), BankAccount (AccountNumber), Supplier (SupplierID), Items (itemID), Order (orderID)and Bill (BillId).<br><br>Each table also includes the original attributes of the entity as columns.<br>For example, the Customer table includes UserID, Password, etc.</td></tr>
<tr><td>Step2: Mapping of Weak Entity Types</td><td>We don’t have any weak entities in our EER.</td></tr>
<tr><td>Step 3: Mapping of Binary 1:1 Relation Types</td><td>There is a one-to-one relationship between Order and Bill, and both entities participate totally in the relationship.<br><br>Due to this full participation on each side, we combined the Order and the Bill tables into a new table named OrderWithBill.<br><br>We take the primary keys OrderID and BillID, which together uniquely represent the relationship between an order and its bill.<br>We also Discount as attributes in the merged table.<br>This means that each bill is associated with a single order, and each order is associated with a single bill.<br><br>All the related data is now contained in the OrderWithBill table, and there is no longer an Order table or a Bill table.</td></tr>
<tr><td>Step4: Mapping of Binary 1:N Relationship Types</td><td>We have more than one one-to-many relationship in our EER.<br><br>The first one is between the Customer and the BankAccount. We take the customerID (PK) and add it to the BankAccount table as a foreign key.<br><br>The second relationship is between the Customer and the Order. Also here, we add the customerID into the order table as a foreign key.<br><br>The last one is between the BankAccount and the Bill, so we take the account number and put it in the Bill table as a foreign key.<br><br>This way, each one-to-many relationship is handled by placing the primary key of the one side as a foreign key in the many side table.</td></tr>
</table>

| | |
|---|---|
| **Step5: Mapping of Binary M:N Relationship Types.** | The M:N relationships type Exists is mapped by creating a relation **Exists** in the relational database schema. The primary keys of the **Order and Item** relations are included as foreign keys in Exists.<br>The M:N relationship type **Supply** is mapped by creating a relation Supply in the relational schema. The primary keys of the **Supplier and Item** relations are included as foreign keys in Supply. |

**Steps Not applied in our EER:**

From Step 6 to Step 9, not all of them apply to our diagram.

For example, we don't have any multivalued attributes, or relationships with more than two entities.

Also, our model doesn't include specialization, generalization, or union types.

So we didn't use those steps because they don't match our EER.

## 4. Conclusion

In summary, this project has clarified and the problems that could face Online Retail Application if it is not designed efficiently, and outlined a well-designed database system.

The system successfully enables essential features such as customer sign-up, secure management of multiple bank accounts, buying products in different quantities, and adaptive billing that accounts for discounts.

With a well-structured EER diagram and its related relational schema, we have illustrated how various entities such as customers, bank accounts, products, orders, and suppliers interact with each other in the system.

The design guarantees data integrity, safety, and scalability, facilitating accurate purchase tracking and efficient supplier management.

The proposed database system lays a solid groundwork for overseeing retail transactions in an online setting, improving both user satisfaction and operational effectiveness.